Table 1: Summary of Key Notations

Notation	Description
System Foundation	
λ, n, Δ	Security parameter, RLWE dimension, Scaling factor
$\mathbb{Z}_q, \mathcal{R}_q, \mathcal{R}_p$	Integer, ciphertext, and plaintext rings
φ, M	Ring isomorphism; Prime bounding modulus
χ_s, χ_e	Secret key and Gaussian error distributions
Client & Training Context	
$\mathcal{C}, N, K$	All client set, total count, selected count
$T, t, \mathcal{E}$	Total rounds, current round, local epochs
J, B, j	Total packs, PRPG seed, pack index
$R, sd_{i,j}$	Prefetch window; Pairwise PRG seed
$\mathcal{S}^{(t)}, \bar{\mathcal{S}}^{(t)}, \mathcal{P}^{(t^*)}$	Active, inactive, prefetched client sets
S, s_i	Global decryption key; Local encryption key
Consensus & Model Updates	
$\mathcal{T}_i^{(t)}, \mathcal{K}^{(t)}$	Local Top-k indices; Consensus mask
$\mathbf{v}_i^{(t)}, v_{i,j}^{(t)}$	Vote bitmap; Binary voting element
$s_j^{(t)}, \theta, \tau_\theta$	Global score; Selection ratio; Truncation threshold
$\mathbf{W}^{(t)}, \mathbf{w}_i^{(t)}$	Global model; Local model
Gradients & Residuals	
$\mathbf{g}_i^{(t)}, \tilde{\mathbf{g}}_i^{(t)}, \mathbf{r}_i^{(t)}$	Raw gradient; Compensated gradient; Pack Residual
$\mathrm{pt}_{i,j}^{(t)}, e_{i,j}^{(t)}$	Encoded gradient polynomial; Noise polynomial
γ_d, γ_s	Per-round and cross-round decay factors
Privacy & Cryptography	
$\mathbf{y}_{i,j}^{(t)}, \mathbf{z}_i^{(t)}, \hat{\mathbf{v}}_i^{(t)}$	Pairwise vector; Perturbation; Obfuscated vote
$\mathsf{ct}_{i,j}^{(t)}, \mathsf{ct}_{k,j}^{(t)}$	Local upload and compensatory ciphertexts
$C^{(t)}, C_j^{(t)}$	Global and pack-level aggregated ciphertexts; $C_j^{(t)}$ includes active gradients and KGC compensation
$\mathbf{\Delta}^{(t)}$	KGC compensatory perturbation
Prefetching & Aggregation	
$\mathcal{Q}$	Sliding queue of the latest R rounds' encrypted updates
$\mathcal{L}_{i,j}$	Client i's prefetch accumulator for pack j
$t_{\mathrm{last},i}$	Last training round of client i
$\mathbf{m}_{\mathrm{pre}}$	Prefetched plaintext from local accumulators
α, β	EMA decay and over-commitment factors
$\hat{d}^{(t)}, d^{(t)}$	EMA estimate and measured wall-clock round duration

1 Updated Notation Table

2 Consensus Selection Strategies

Let $\bar{\mathbf{v}}^{(t)}$ denote the global vote tally vector, where each entry

$$\bar{v}_j^{(t)} = \sum_{i \in \mathcal{S}^{(t)}} v_{i,j}^{(t)} \tag{1}$$

counts how many active clients voted for pack j at round t. The server obtains the consensus set $\mathcal{K}^{(t)}$ using one of the following rules.

- **Union (ECPFL-UN):** retains every pack that receives at least one vote,

$$\mathcal{K}^{(t)} = \left\{ j \in [J] \;\middle|\; \bar{v}_j^{(t)} \geq 1 \right\}. \tag{2}$$

- **Rank (ECPFL-RK-θ):** sorts packs by their global vote tallies in descending order and retains the top-$\lfloor \theta J \rfloor$ packs, where $\theta \in (0, 1]$,

$$\mathcal{K}^{(t)} = \left\{ \pi_\ell \in [J] \;\middle|\; 1 \leq \ell \leq \lfloor \theta J \rfloor,\ \bar{v}_{\pi_1}^{(t)} \geq \bar{v}_{\pi_2}^{(t)} \geq \cdots \geq \bar{v}_{\pi_J}^{(t)} \right\}. \tag{3}$$

- **Threshold (ECPFL-TH-θ):** retains a pack only if its accumulated votes reach a proportion $\theta \in (0, 1]$ of the active clients,

$$\mathcal{K}^{(t)} = \left\{ j \in [J] \;\middle|\; \bar{v}_j^{(t)} \geq \theta \cdot |\mathcal{S}^{(t)}| \right\}. \tag{4}$$